CS3236 Information Theory

AY 25/26 Sem 2 — github/omgeta

1. Introduction

Data Compression

Settings:

i. Lossless: recover data exactly (e.g. `.zip`)
   - L bits for N distinct messages: $L \geq \log_2 N$

ii. Lossy: allow distortion for smaller size (e.g. `.jpeg`)

Approaches:

i. Variable-length: shorter codewords for higher-probability sequences

ii. Fixed-length: encode "typical" sequences; small failure probability for low-probability sequences

Source Coding Theorem (informal):

i. Average lossless compression $\geq H$ (entropy)

Data Communication

Binary Symmetric Channel: bits flip indep. w.p. δ

$$\mathbb{P}[\text{bit flip}] = \delta, \quad \mathbb{P}[\text{no flip}] = 1-\delta, \quad \delta \in (0, \frac{1}{2})$$

Approaches:

i. Uncoded:
   - $\mathbb{P}[\text{correct}] = (1-\delta)^n$
   - $R = 1$

ii. Repetition Code (odd $n, 1$):
repeat bits n times, decide by majority vote
   - $\mathbb{P}[\text{correct}] = \sum_{i=0}^{(n-1)/2} \binom{n}{i} \delta^i (1-\delta)^{n-i}$
   - $R = \frac{1}{n}$

iii. Hamming Code $(7, 4)$: 3 parity bits for 4 data bits (detect ≤ 2 errors, correct 1)
   - $\mathbb{P}[\text{correct}] = (1-\delta)^7 + 7\delta(1-\delta)^6$
   - Rate $R = \frac{4}{7}$

Channel Coding Theorem (informal):

i. If $R < C$ (capacity), $\exists$ codes s.t. $\mathbb{P}[\text{error}] \to 0$

2. Measures

Information

Information from event A w.p. p: $\psi(A) = \log(\frac{1}{p})$

i. Uncertainty of/Info gained from A occuring

Entropy

Entropy of X over $\mathcal{X}$:

$$H(X) = \mathbb{E}_{X \sim P_X}\left[\log_2 \frac{1}{P_X(X)}\right]$$

$$H_2(p) = p \log_2 \frac{1}{p} + (1-p) \log_2 \frac{1}{1-p}$$

i. $H(X) \geq 0$
with equality iff X deterministic (Non-negativity)

ii. $H(X) \leq \log_2 |\mathcal{X}|$
with equality iff $X \sim \text{Unif}(\mathcal{X})$ (Upper bound)

iii. $H(f(X)) \leq H(X)$
with equality iff f is one-to-one (Function)

Conditional Entropy of X, Y:

$$H(Y|X) = \mathbb{E}_{(X,Y) \sim P_{XY}}\left[\log_2 \frac{1}{P_{Y|X}(Y|X)}\right]$$
$$= \sum_x P_X(x) H(Y|X=x)$$

1. $H(Y|X) \leq H(Y)$
with equality iff $X \perp Y$ (Conditioning $\downarrow$)
2. If Y depends on X only through $f(X)$,
$H(Y|X) = H(Y|f(X))$ (Info-Preserving)

Joint Entropy of X, Y:

$$H(X,Y) = \mathbb{E}_{(X,Y) \sim P_{XY}}\left[\log_2 \frac{1}{P_{XY}(X,Y)}\right]$$

1. $H(X,Y) = H(Y,X)$ (Symmetric)
2. $H(X,Y) = H(X) + H(Y|X)$
$H(X_1, \ldots, X_n) = \sum_{i=1}^n H(X_i|X_1, \ldots, X_{i-1})$(Chain)
   - $H(X,Y|Z) = H(X|Z) + H(Y|X,Z)$
3. $H(X_1, \ldots, X_n) \leq \sum_{i=1}^n H(X_i)$
with equality iff $X_1, \ldots, X_n$ indep. (Sub-additivity)

Relative Entropy (KL Divergence)

Relative entropy of PMFs P_X, Q_X:

$$D(P\|Q) = \mathbb{E}_{X \sim P}\left[\log_2 \frac{P(X)}{Q(X)}\right]$$

i. $D(P\|Q) \geq 0$
with equality iff $P = Q$ (Non-negativity)

ii. $D(P_{XY}\|Q_{XY}) = D(P_X\|Q_X) + D(P_{Y|X}\|Q_{Y|X}|P_X)$ (Chain)

iii. P_Y, Q_Y from applying $V(y|x)$ to P_X, Q_X
$\implies D(P_Y\|Q_Y) \leq D(P_X\|Q_X)$ (DPI)

Mutual Information

Mutual Information of X, Y over $\mathcal{X}, \mathcal{Y}$:

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) = H(X) - H(X|Y) \\ &= H(X) + H(Y) - H(X,Y) \\ &= D(P_{XY}\|P_X \times P_Y) \end{aligned}$$

i. $I(X;Y) = I(Y;X)$ (Symmetric)

ii. $I(X;Y) \geq 0$
with equality iff $X \perp Y$ (Non-negativity)

iii. $I(X;Y) \leq H(X) \leq \log_2 |\mathcal{X}|$
$I(X;Y) \leq H(Y) \leq \log_2 |\mathcal{Y}|$ (Upper bounds)

Conditional Mutual Information for X, Y, Z:

$$I(X;Y|Z) = H(Y|Z) - H(Y|X,Z)$$

i. $I(X;Y|Z) \geq 0$
with equality iff $X \perp Y \mid Z$ (Non-negativity)

Joint Mutual Information for X, Y, Z:

$$\begin{aligned} I(X,Y;Z) &= H(Z) - H(Z|X,Y) \\ &= I(X;Z) + I(Y;Z|X) \\ I(X;Y,Z) &= I(X;Y) + I(X;Z|Y) \end{aligned}$$

i. $I(X_1, \ldots, X_n; Y) = \sum_{i=1}^n I(X_i; Y|X_1, \ldots, X_{i-1})$ (Chain)

ii. $X \to Y \to Z \Rightarrow I(X;Z) \leq I(X;Y)$
with equality iff $I(X;Y|Z) = 0$ (DPI)

iii. If $(Y_i \perp Y_j) \mid (X_1, \ldots, X_n) \wedge Y_i$ depends only on X_i,
$\implies I(X_1, \ldots, X_n; Y_1, \ldots, Y_n) \leq \sum_{i=1}^n I(X_i; Y_i)$ (Partial Sub-additivity)

3. Symbol-Wise Source Coding

Symbol-wise source coding maps each $x \in \mathcal{X}$ to some binary sequence $C(x)$ with length $\ell(x)$

$$L(C) = \mathbb{E}[\ell(X)] = \sum_{x \in \mathcal{X}} P_X(x)\ell(x)$$

Decodability Conditions:

i. Nonsingular: $x \neq x' \Rightarrow C(x) \neq C(x')$

ii. Uniquely Decodable:
$C(x_1) \cdots C(x_n)$ decodes to a unique $x_1, \ldots, x_n$
- $\sum_{x \in \mathcal{X}} 2^{-\ell(x)} \leq 1$ (Kraft's Inequality)

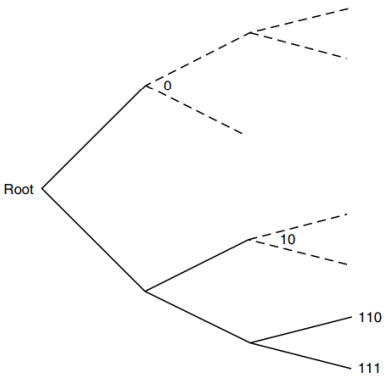
Prefix-free Code

Prefix-free codes have no codeword as prefix of another:

+ Uniquely decodable
+ Loses no optimality in average length
+ Instantaneous left-to-right decoding with greedy algorithm; termination uniquely identifies codeword

i. Existence: $\{\ell(x)\}_{x \in \mathcal{X}}$ satisfy Kraft's inequality $\iff \exists$ a prefix-free code mapping $x \in \mathcal{X}$ to codeword of length $\ell(x)$

ii. $L(C) \geq H(X)$
with equality iff $P_X(x) = 2^{-\ell(x)}$ (Entropy-Bound)
- Ideal length $\ell^*(x) = \log_2 \frac{1}{P_X(x)}$



Shannon–Fano Code

Shannon–Fano code rounds ideal lengths up to the nearest integer:

$$\ell(x) = \lceil \log_2 \frac{1}{P_X(x)} \rceil$$

i. $\sum_x 2^{-\ell(x)} \leq 1$ so $\exists$ prefix-free code with lengths

ii. $\log_2 \frac{1}{P_X(x)} \leq \ell(x) < \log_2 \frac{1}{P_X(x)} + 1$
$\implies H(X) \leq L(C) < H(X) + 1$

iii. Mismatch (true P_X but $\ell(x)$ uses Q_X):
$H(X) + D(P_X \| Q_X) \leq L(C) < H(X) + D(P_X \| Q_X) + 1$

Huffman Code

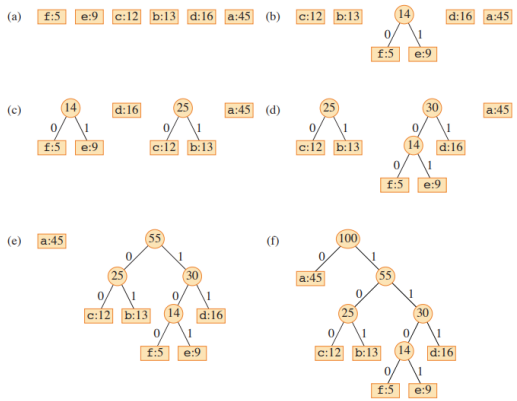
Huffman code is the optimal prefix-free symbol code minimizing $L(C)$:

i. Optimality:
for any prefix-free code C, $L(C_{\text{Huff}}) \leq L(C)$
- $H(X) \leq L(C_{\text{Huff}}) < H(X) + 1$
- $\sum_x 2^{-\ell(x)} = 1$ from optimality

Algorithm:

i. Construct by repeatedly merging the two least-probable nodes into one with probability sum

ii. Label merge edges 0/1; codeword is the root-to-leaf path for each symbol



Interpretation as Yes/No Questions

A sequence of yes/no questions that uniquely identifies $x \in \mathcal{X}$ is equivalent to a binary code:

i. Each symbol x has a distinct sequence of answers, hence a distinct binary codeword

ii. If symbol x is identified after $\ell(x)$ questions, then its codeword has length $\ell(x)$

iii. Conversely, a binary code induces a questioning scheme by asking for successive codeword bits

iv. Hence the average number of yes/no questions is

$$H(X) \leq \mathbb{E}[Q] = L(C) < H(X) + 1$$

v. Therefore, Huffman coding gives an optimal yes/no questioning scheme

Memory Limitation

Symbol-wise codes ignore dependence across symbols.

Mitigate via **block coding**:
group $\mathcal{X}^N$ and code w.r.t. $P_{X_1,\ldots,X_N}$:

+ Exploit sequence probabilities ("apple" $\gg$ "ealpp")
+ If i.i.d, $H(X_1, \ldots, X_N) = \sum_{i=1}^N H(X_i)$
$\Rightarrow NH(X) \leq L(C) \leq NH(X) + 1$
$\Rightarrow H(X) \leq$ Avg. length per symbol $\leq H(X) + \frac{1}{N}$

– Determining $P_{X_1,\ldots,X_N}$ accurately is hard
– Complexity grows exponentially: $|\mathcal{X}|^N$ symbols (sorting / Huffman)

4. Block Source Coding

Block source coding maps $\mathbf{X} \in \mathcal{X}^n$ to $m \in \{1, \ldots, M\}$:

$$R = \frac{1}{n} \log_2 M$$

i. Finite $|\mathcal{X}|$ (Discrete)
ii. $P_{\mathbf{X}}(\mathbf{x}) = \prod P_X(x_i)$ (Memoryless)
iii. Codebook known at encoder and decoder; only encoder knows m
iv. Error when $M < |\mathcal{X}|^n$ with $P_e = \mathbb{P}[\hat{\mathbf{X}} \neq \mathbf{X}]$

Source Coding Theorem:

i. $R > H(X) \Rightarrow \forall \epsilon > 0\ \exists n\ P_e \leq \epsilon$ (Achievability)
ii. $R < H(X) \Rightarrow \exists \epsilon > 0\ \forall n\ P_e > \epsilon$ (Converse)

Typical Sets

Typical Set for $\epsilon > 0$:

i. $\mathbb{P}[\mathbf{X} \in T_n(\epsilon)] \to 1$ (High Prob.)
   - $\frac{1}{n} \sum \log_2 \frac{1}{P_X(X_i)} \to H(X)$ by LLN since $\mathbb{E}[\log_2 \frac{1}{P_X(X)}] = H(X)$
ii. $|T_n(\epsilon)| \leq 2^{n(H(X)+\epsilon)}$ (Upper)
   - $1 \geq \mathbb{P}[\mathbf{X} \in T_n(\epsilon)] \geq |T_n(\epsilon)| \cdot 2^{-n(H(X)+\epsilon)}$
iii. $|T_n(\epsilon)| \geq (1 - o(1)) 2^{n(H(X)-\epsilon)}$ (Lower)
   - $1 - o(1) = \mathbb{P}[\mathbf{X} \in T_n(\epsilon)] \leq |T_n(\epsilon)| \cdot 2^{-n(H(X)-\epsilon)}$
iv. $\sim 2^{nH(X)}$ typical seqs, each w.p. $\sim 2^{-nH(X)}$ (AEP)

Coding Scheme (map non-typical to dummy):

i. $M = |T_n(\epsilon)| + 1 \leq 2^{n(H(X)+\epsilon)} + 1$
ii. $R = \frac{1}{n} \log_2 M \leq H(X) + \epsilon + o(1)$
iii. $P_e \leq \mathbb{P}[\mathbf{X} \notin T_n(\epsilon)] \to 0$
iv. $\epsilon \to 0 \Rightarrow R \to H(X),\ P_e \to 0$

Fano's Inequality

Fano's Inequality for $X, \hat{X}$ on $\mathcal{X}$:

$$\begin{aligned} H(X|\hat{X}) &\leq H_2(P_e) + P_e \log_2(|\mathcal{X}| - 1) \\ &\leq 1 + P_e \log_2 |\mathcal{X}| \end{aligned}$$

Derivation via error indicator $E = \mathbf{1}\{X \neq \hat{X}\}$:

i. $H(X|\hat{X}) = H(X, E|\hat{X})$ (E det. fn of $X, \hat{X}$)
ii. $= H(E|\hat{X}) + H(X|\hat{X}, E) \leq H(E) + H(X|\hat{X}, E)$ (Chain)
iii. $H(E) = H_2(P_e)$; $H(X|\hat{X}, E = 0) = 0$; $H(X|\hat{X}, E = 1) \leq \log_2(|\mathcal{X}| - 1)$

Applying to $(\mathbf{X}, \hat{\mathbf{X}})$ on $\mathcal{X}^n$:

i. $H(\mathbf{X}|\hat{\mathbf{X}}) \leq 1 + P_e \cdot n \log_2 |\mathcal{X}|$
ii. $I(\mathbf{X}; \hat{\mathbf{X}}) = H(\mathbf{X}) - H(\mathbf{X}|\hat{\mathbf{X}}) \geq nH(X) - P_e \cdot n \log_2 |\mathcal{X}| - 1$
iii. $I(\mathbf{X}; \hat{\mathbf{X}}) \leq H(\hat{\mathbf{X}}) \leq nR$ ($\hat{\mathbf{X}}$ takes $M = 2^{nR}$ values)
iv. $P_e \geq \frac{1}{\log_2 |\mathcal{X}|} \left(H(X) - R - \frac{1}{n}\right)$
v. $R < H(X) \Rightarrow P_e \not\to 0$ (Impossibility)

5. Channel Coding

Channel coding transmits $m \in \{1, \dots, M\}$ over $P_{Y|X}$ via codebook $\mathcal{C} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(M)}\}$:

$$R = \tfrac{1}{n} \log_2 M \iff M = 2^{nR}$$

i. Finite $|\mathcal{X}|, |\mathcal{Y}|$ (Discrete)
ii. $P_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}|\mathbf{x}) = \prod P_{Y|X}(y_i|x_i)$ (Memoryless)
iii. Error with $P_e = \mathbb{P}[\hat{m} \neq m]$

Channel Capacity:

$$C = \max_{P_X} I(X;Y)$$

i. $R < C \Rightarrow \forall \epsilon > 0\ \exists n\ P_e \leq \epsilon$ (Achievability)
ii. $R > C \Rightarrow \exists \epsilon > 0\ \forall n\ P_e > \epsilon$ (Converse)

Standard Channels

Assuming uniform P_X:

i. Noiseless ($\mathcal{X} = \mathcal{Y} = \{0,1\}$, $Y = X$):
   - $I(X;Y) = H(X) - H(X|Y) = H(X)$
   - $C = \max_{P_X} H(X) = \log_2 |\mathcal{X}| = 1$
ii. Binary Symmetric
   ($\mathcal{X} = \mathcal{Y} = \{0,1\}$, flip w.p. δ):
   - By symmetry, $H(Y|X = x) = H_2(\delta)$
   - By indep., $H(Y|X) = \sum_x P_X(x) H_2(\delta) = H_2(\delta)$
   - $I(X;Y) = H(Y) - H_2(\delta) \leq 1 - H_2(\delta)$
   - $C = 1 - H_2(\delta)$
iii. Binary Erasure
   ($\mathcal{X} = \{0,1\}$, $\mathcal{Y} = \{0,1,e\}$, erase w.p. ϵ):
   - $Y \in \{0,1\}$ fully determines X
     $\Rightarrow H(X|Y \neq e) = 0$
   - $Y = e$ leaves X unchanged
     $\Rightarrow H(X|Y = e) = H(X)$
   - $H(X|Y) = P_Y(e) H(X|Y = e) = \epsilon H(X)$
   - $I(X;Y) = H(X) - \epsilon H(X) = (1-\epsilon) H(X)$
   - $C = (1-\epsilon) \max_{P_X} H(X) = 1 - \epsilon$

Jointly Typical Sets

Jointly Typical Set for $\epsilon > 0$:

$$\begin{aligned} 2^{-n(H(X)+\epsilon)} &\leq P_{\mathbf{X}}(\mathbf{x}) \leq 2^{-n(H(X)-\epsilon)} \\ 2^{-n(H(Y)+\epsilon)} &\leq P_{\mathbf{Y}}(\mathbf{y}) \leq 2^{-n(H(Y)-\epsilon)} \\ 2^{-n(H(X,Y)+\epsilon)} &\leq P_{\mathbf{XY}}(\mathbf{x},\mathbf{y}) \leq 2^{-n(H(X,Y)-\epsilon)} \end{aligned}$$

i. $\mathbb{P}[(\mathbf{X},\mathbf{Y}) \in T_n(\epsilon)] \to 1$ (High Prob.)
ii. $|T_n(\epsilon)| \leq 2^{n(H(X,Y)+\epsilon)}$ (Upper Bound)
iii. $(\mathbf{X}', \mathbf{Y}') \sim P_{\mathbf{X}} \times P_{\mathbf{Y}}$ indep.
   $\Rightarrow \mathbb{P}[(\mathbf{X}', \mathbf{Y}') \in T_n(\epsilon)] \leq 2^{-n(I(X;Y)-3\epsilon)}$ (Indep.)

Random Coding Scheme (generate $X_i^{(m)} \overset{\text{i.i.d.}}{\sim} P_X$, decode to unique $\tilde{m}$ with $(\mathbf{x}^{(\tilde{m})}, \mathbf{y}) \in T_n(\epsilon)$):

i. Sent: $(\mathbf{X}^{(m)}, \mathbf{Y}) \sim P_{XY}^n$;
   Other: $(\mathbf{X}^{(\tilde{m})}, \mathbf{Y}) \sim P_{\mathbf{X}} \times P_{\mathbf{Y}}$
ii. $P_e^{(m)} \leq \mathbb{P}[(\mathbf{X}^{(m)}, \mathbf{Y}) \notin T_n(\epsilon)]$
   $+ \sum_{\tilde{m} \neq m} \mathbb{P}[(\mathbf{X}^{(\tilde{m})}, \mathbf{Y}) \in T_n(\epsilon)]$ (Union)
iii. $P_e^{(m)} \leq \delta_n + M\, 2^{-n(I(X;Y)-3\epsilon)}$
   $= \delta_n + 2^{-n(I(X;Y)-R-3\epsilon)}$
iv. $R < I(X;Y) - 3\epsilon \Rightarrow P_e \to 0$; best code $\geq$ average
v. Max over $P_X \Rightarrow R \to C$ (Existence only)

Fano-Based Converse

Applying Fano to $(m, \hat{m})$ with $H(m) = \log_2 M$ uniformly:

i. $H(m|\hat{m}) \leq 1 + P_e \log_2 M$
ii. $I(m;\hat{m}) = H(m) - H(m|\hat{m}) \geq (1 - P_e) \log_2 M - 1$
iii. $I(m;\hat{m}) \overset{\text{DPI}}{\leq} I(\mathbf{X};\mathbf{Y}) \leq \sum_i I(X_i;Y_i) \leq nC$
   - $m \to \mathbf{X} \to \mathbf{Y} \to \hat{m}$ (DPI)
   - $H(\mathbf{Y}) \leq \sum_i H(Y_i)$ (Sub-additivity)
   - $H(\mathbf{Y}|\mathbf{X}) = \sum_i H(Y_i|X_i)$ (Memoryless)
   - $I(\mathbf{X};\mathbf{Y}) \leq \sum_i I(X_i;Y_i) \leq nC$ (Capacity)
iv. $P_e \geq 1 - \frac{C + 1/n}{R} \Rightarrow R > C$ bounds P_e away from 0

Useful MI Laws

Conditional MI decomposition:

$$I(X;U|V) = \sum_v P_V(v)\, I(X;U|V = v)$$

i. V determines U or $X \Rightarrow I(X;U|V = v) = 0$

Chain Rule (multi-output):

$$\begin{aligned} I(X;U,V) &= I(X;V) + I(X;U|V) \\ &= I(X;U) + I(X;V|U) \end{aligned}$$

Deterministic Functions:

i. f deterministic: $I(X; f(U,V)) \leq I(X;U,V)$ (DPI)
ii. f one-to-one on support:
   $I(X; f(U,V)) = I(X;U,V)$ (Invariance)

Zero-Capacity Condition:

i. $P_{Y|X}(\cdot|x)$ same
   $\forall x \Rightarrow Y \perp X \Rightarrow I(X;Y) = 0 \Rightarrow C = 0$
ii. Check: $\mathbb{P}[Y = y|X = 0] = \mathbb{P}[Y = y|X = 1]\ \forall y$

6. Continuous-Alphabet Channels

Differential Entropy

Differential Entropy of continuous X with PDF f_X:

$$\begin{aligned} h(X) &= \mathbb{E}_{X\sim f_X}\left[\log_2 \tfrac{1}{f_X(X)}\right] \\ &= \int_{\mathbb{R}} f_X(x)\log_2 \tfrac{1}{f_X(x)}\,dx \end{aligned}$$

i. $h(X) \le \frac{1}{2}\log_2(2\pi e \operatorname{Var}[X])$
with equality iff X Gaussian (Upper bound)

ii. $h(X+c) = h(X)$ (Shift-invariance)

iii. Preserved: chain rule, conditioning $\downarrow$, sub-additivity

Lost:

i. $h(X) \in \mathbb{R}$ may be negative (Non-negativity)

ii. $h(cX) = h(X) + \log_2|c|$ (Invariance)

Key Distributions:

i. $X \sim \operatorname{Unif}(a,b)$: $h(X) = \log_2(b-a)$

ii. $X \sim \mathcal{N}(\mu,\sigma^2)$: $h(X) = \frac{1}{2}\log_2(2\pi e\sigma^2)$

Continuous Relative Entropy

Relative entropy of PDFs f_X, g_X:

$$\begin{aligned} D(f\|g) &= \mathbb{E}_{X\sim f}\left[\log_2 \tfrac{f(X)}{g(X)}\right] \\ &= \int_{\mathbb{R}} f(x)\log_2 \tfrac{f(x)}{g(x)}\,dx \end{aligned}$$

i. Preserved: non-negativity, chain rule, DPI

Continuous Mutual Information

Mutual Information of continuous X, Y:

$$\begin{aligned} I(X;Y) &= h(Y) - h(Y|X) = h(X) - h(X|Y) \\ &= h(X) + h(Y) - h(X,Y) \\ &= D(f_{XY}\|f_X \times f_Y) \end{aligned}$$

i. Preserved: non-negativity, chain rule, DPI

ii. $I(X;Y) = I(\phi(X);\psi(Y))$ for invertible ϕ,ψ
(Invariance)

Gaussian Channel

Additive noise channel with $Z \sim \mathcal{N}(0,\sigma^2) \perp X$:

$$Y = X + Z \quad\Longrightarrow\quad f_{Y|X}(y|x) = f_Z(y-x)$$

i. Unconstrained $X \Rightarrow C = \infty$

ii. Power constraint: $\mathbb{E}[X^2] \le P$

Gaussian Channel Capacity:

$$C(P) = \max_{f_X:\,\mathbb{E}[X^2]\le P} I(X;Y) = \tfrac{1}{2}\log_2\!\left(1+\tfrac{P}{\sigma^2}\right)$$

i. $h(Z) = \frac{1}{2}\log_2(2\pi e\sigma^2)$ (Gaussian)

ii. $\operatorname{Var}[Y] = \operatorname{Var}[X] + \operatorname{Var}[Z] \le P + \sigma^2$
$\Longrightarrow h(Y) \le \frac{1}{2}\log_2(2\pi e(P+\sigma^2))$ (Max-Entropy)

iii. $I(X;Y) = h(Y) - h(Y|X) = h(Y) - h(Z)$ $(Z \perp X)$
$\le \frac{1}{2}\log_2(1+P/\sigma^2)$

iv. Tight when $X \sim \mathcal{N}(0,P) \Rightarrow Y \sim \mathcal{N}(0,P+\sigma^2)$

Properties:

i. $C(0) = 0$

ii. $C(P) \to \infty$ as $P \to \infty$

iii. Depends only on signal-to-noise ratio P/σ^2

iv. Low ratio: $C(P) \approx \frac{P}{2\sigma^2\ln 2}$
doubling P doubles C (linear)

v. High ratio: $C(P) \approx \frac{1}{2}\log_2\frac{P}{\sigma^2}$
doubling P adds $\frac{1}{2}$ (diminishing)

7. Practical Channel Codes

Linear Codes

Linear codes map $\mathbf{u} \in \{0,1\}^k$ to $\mathbf{x} \in \{0,1\}^n$ via modulo-2, rate $R = k/n$:

$$\mathbf{x} = \mathbf{u}G \pmod 2 \iff x_i = \bigoplus_{j=1}^{k} u_j g_{j,i}$$

i. G is $k \times n$ generator matrix;
$g_{j,i} = 1$ iff u_j in parity check at position i
ii. Linear: $\mathbf{u}G \oplus \mathbf{u}'G = (\mathbf{u} \oplus \mathbf{u}')G$
sum of codewords is a codeword
iii. General: all n bits are arbitrary parity checks of $\mathbf{u}$
iv. Systematic: $G = [I_k\ P] \Rightarrow$ first k bits of $\mathbf{x}$ equal $\mathbf{u}$
v. Storage: nk bits vs. exponential generic
vi. Linearly independent rows allow unique decoding

Example matrices:

$$G_{\text{par}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}$$

$$G_{\text{Ham}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}$$

Parity-Check Matrix

$n \times (n-k)$ matrix H satisfying:

$$\mathbf{x}H = 0 \iff \mathbf{x} \text{ is a valid codeword}$$

Systematic construction:

$$G = [I_k\ P] \Rightarrow H = \begin{pmatrix} P \\ I_{n-k} \end{pmatrix}$$

Example matrices:

$$H_{\text{par}} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$H_{\text{Ham}} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

BSC output $\mathbf{y} = \mathbf{x} \oplus \mathbf{z}$ gives syndrome:

$$S = \mathbf{y}H = (\mathbf{x} \oplus \mathbf{z})H = \mathbf{x}H \oplus \mathbf{z}H = \mathbf{z}H$$

i. Depends only on error $\mathbf{z}$, not on $\mathbf{x}$
ii. Single flip at position $i \iff S = i$-th row of H
iii. Distinct nonzero rows $\Rightarrow$ every single-bit error correctable

Distance Properties

Hamming distance and weight:

$$d_H(\mathbf{x}, \mathbf{x}') = \sum_i \mathbf{1}\{x_i \neq x_i'\}, \quad w(\mathbf{x}) = \sum_i \mathbf{1}\{x_i = 1\}$$

Minimum distance of codebook $\mathcal{C}$:

$$d_{\min} = \min_{\mathbf{x} \neq \mathbf{x}' \in \mathcal{C}} d_H(\mathbf{x}, \mathbf{x}')$$

i. Linear code: $d_{\min} = \min_{\mathbf{x} \neq 0 \in \mathcal{C}} w(\mathbf{x})$ (Weight)
- $\mathbf{x} \oplus \mathbf{x}'$ is a codeword with weight $d_H(\mathbf{x}, \mathbf{x}')$

ii. Correct up to $d_{\min} - 1$ erasures
iii. Correct up to $\lfloor \frac{d_{\min}-1}{2} \rfloor$ bit flips
non-overlapping balls of radius $\frac{d_{\min}-1}{2}$

Minimum Distance Decoding

Optimal decoder minimizing P_e under uniform m:

$$\hat{m} = \arg \max_{j=1,\ldots,M} P_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}|\mathbf{x}^{(j)})$$

i. Equivalent to maximum-a-posteriori under uniform prior (Bayes)
ii. BSC ($\delta < \frac{1}{2}$):
$\log P_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}|\mathbf{x}) = n \log(1-\delta) - d_H(\mathbf{x}, \mathbf{y}) \log \frac{1-\delta}{\delta}$
iii. $\Rightarrow \hat{m} = \arg\min_j d_H(\mathbf{x}^{(j)}, \mathbf{y})$ (Min-distance)

Syndrome Decoding (linear code on BSC):

i. Compute syndrome $S = \mathbf{y}H$
ii. Find $\hat{\mathbf{z}} = \arg\min_{\tilde{\mathbf{z}}:\, \tilde{\mathbf{z}}H = S} w(\tilde{\mathbf{z}})$
iii. Output $\hat{\mathbf{x}} = \mathbf{y} \oplus \hat{\mathbf{z}}$; first k bits give $\hat{\mathbf{u}}$ (systematic)
iv. Search: try $\tilde{\mathbf{z}}$ of weight $0, 1, 2, \ldots$ until $\tilde{\mathbf{z}}H = S$

\+ Fast when few errors occur

− Exponential worst-case complexity

Appendix

Random Variables

Probability Mass Function (PMF) for discrete X:

$$P_X(x) = \mathbb{P}[X = x]$$

i. $\forall x,\ 0 \le P_X(x) \le 1$
ii. $\sum_x P_X(x) = 1$

Probability Density Function (PDF) for continuous X:

$$\int_a^b f_X(x)\, dx = \mathbb{P}[a \le X \le b]$$

i. $\forall x,\ f_X(x) \ge 0$
ii. $\int_x f_X(x)\, dx = 1$

Expectation

Expectation of X:

$$\mathbb{E}[X] = \sum_x P_X(x)\, x$$

i. $\mathbb{E}[f(X)] = \sum_x P_X(x)\, f(x)$ (Function)
ii. $\mathbb{E}[c(X)] = c\mathbb{E}[X]$ (Scaling)
iii. $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ (LoE)
iv. $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y] \quad \text{if } X \perp Y$ (Product)
v. $\mathbb{E}[\mathbf{1}\{A\}] = \mathbb{P}[A]$ (Indicator)

Variance

Variance of X:

$$\begin{aligned}\mathrm{Var}[X] &= \mathbb{E}[(X-\mu)^2] \\ &= \mathbb{E}[X^2] - \mu^2\end{aligned}$$

i. $\mathrm{Var}[cX] = c^2\,\mathrm{Var}[X]$ (Scaling)
ii. $\mathrm{Var}[X+Y] = \mathrm{Var}[X] + \mathrm{Var}[Y] + 2\,\mathrm{Cov}[X,Y]$
$= \mathrm{Var}[X] + \mathrm{Var}[Y] \quad \text{if } X \perp Y$ (Sum)
iii. $\mathrm{Cov}[X,Y] = \mathbb{E}[(X-\mu_X)(Y-\mu_Y)]$ (Covariance)

Conditioning

Conditional:

$$P_{Y|X}(y|x) = \frac{P_{XY}(x,y)}{P_X(x)}$$

i. $\mathbb{P}[A] = \sum_x P_X(x)\,\mathbb{P}[A|X=x]$ (Total Probability)
ii. $\mathbb{E}[Y] = \mathbb{E}_X\big[\mathbb{E}_Y[Y|X]\big]$ (Total Expectation)
iii. $\mathbb{P}[A|B] = \frac{\mathbb{P}[A]\,\mathbb{P}[B|A]}{\mathbb{P}[B]}$ (Bayes)

Independence

Independence $X \perp Y$:

$$\begin{aligned}P_{XY}(x,y) &= P_X(x)\,P_Y(y) \\ P_{Y|X}(y|x) &= P_Y(y)\end{aligned}$$

i. $f(X) \perp g(Y)$ if f, g deterministic (Functions)
ii. Joint Indep.: $P_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = \prod_{i=1}^n P_{X_i}(x_i)$
- Pairwise Indep. $\not\Rightarrow$ Joint Indep.

Conditional Independence $X \perp Y \mid Z$:

$$\begin{aligned}P_{XY|Z}(x,y|z) &= P_{X|Z}(x|z)\,P_{Y|Z}(y|z) \\ P_{X|YZ}(x|y,z) &= P_{X|Z}(x|z) \\ P_{Y|XZ}(y|x,z) &= P_{Y|Z}(y|z)\end{aligned}$$

Indicators

Indicator Random Variable:

$$\mathbf{1}\{A\} = \begin{cases} 1, & A \text{ occurs} \\ 0, & A \text{ does not occur}\end{cases}$$

i. $\mathbb{E}[\mathbf{1}\{A\}] = \mathbb{P}[A]$ (Probability)
ii. If $N = \sum_{i=1}^n \mathbf{1}\{A_i\}$, then $\mathbb{E}[N] = \sum_{i=1}^n \mathbb{P}[A_i]$ (Counting)
iii. Uses:
- Counting: count occurences of events
- Simplification: decompose complex RVs into simpler binary indicators

Other

Marginal Distributions:

$$P_X(x) = \sum_y P_{XY}(x,y), \quad P_Y(y) = \sum_x P_{XY}(x,y)$$

Union vs. Intersection:

$$\mathbb{P}[A \cup B] = \mathbb{P}[A] + \mathbb{P}[B] - \mathbb{P}[A \cap B]$$

Union Bound (Boole's Inequality):

$$\mathbb{P}\left[\bigcup_{i=1}^N A_i\right] \le \sum_{i=1}^N \mathbb{P}[A_i]$$

Law of Large Numbers for i.i.d. $X_1, \ldots, X_n$ with mean μ:

$$\mathbb{P}\left[\left|\frac{1}{n}\sum_{i=1}^n X_i - \mu\right| > \epsilon\right] \to 0 \quad \text{as } n \to \infty$$

Jensen's Inequality:

i. If f is convex, then $f(\mathbb{E}[X]) \le \mathbb{E}[f(X)]$
ii. If f is concave, then $f(\mathbb{E}[X]) \ge \mathbb{E}[f(X)]$

Ceiling Bound: $x \le \lceil x \rceil < x + 1$

Logarithm Properties:

i. $\log xy = \log x + \log y$
ii. $\log \frac{1}{x} = -\log x$
iii. $\log x^c = c \log x$
iv. $\log_a x = \frac{\log_b x}{\log_b a}$
v. $\log_e x \le x - 1$ with equality iff $x = 1$
vi. $0 \log \frac{0}{q} = 0$
vii. $p \log \frac{p}{0} = \infty$

Calculus Rules:

i. $\frac{d}{dx} x^c = c x^{c-1}$
ii. $\frac{d}{dx} e^{cx} = c e^{cx}$
iii. $\frac{d}{dx} \ln x = \frac{1}{x}$
iv. $\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$ (Chain)
v. $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$ (Product)
vi. $\frac{d}{dx}\frac{f(x)}{g(x)} = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$ (Quotient)